

Solution Requirements

Date	08 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the application without login in the current version.	Low

2	Authorization Levels	Single-user access is supported; no role-based permissions.	Low
3	External Interfaces	Integrates Speech-to-Text engine and AI semantic similarity model for concept evaluation.	High
4	Transaction Processing	Processes voice input, converts speech to text, analyzes concepts and generates scores.	High
5	Reporting	Displays similarity score, feedback report and performance analysis.	High
6	Business Rules	Only valid audio inputs are processed and evaluated.	High

7	Compliance	Basic user privacy is maintained and no personal voice data is permanently stored.	Medium
8	Other	Responsive Streamlit web interface.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Generate feedback quickly.	Average response time < 5 sec
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2	Scalability	Support multiple users.	Multiple concurrent users
3	Security & Data Privacy	Protect user data and audio.	Secure processing
4	Reliability & Availability	System should remain available.	99% availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to navigate.	Responsive on desktop and mobile
6	Other	Modular and maintainable codebase.	Maintainable project structure